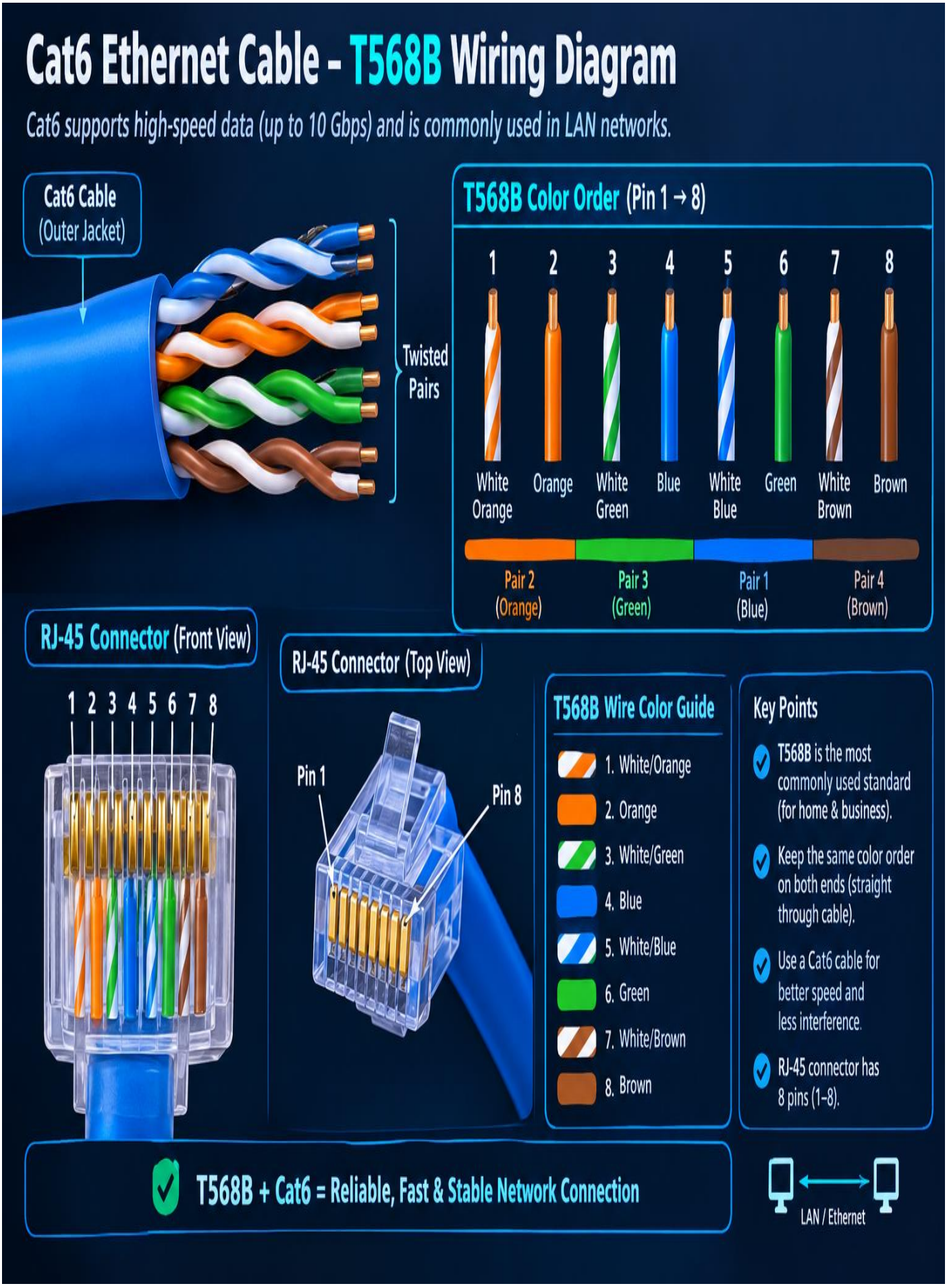
05/10/2026

Assignment-4

Created By
~Pintu mer

1. **Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.**



2. **List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.**

Network Cable	Real-world App/Device	Why it is used
Cat5e	Wi-Fi Router → PC	It is affordable and provides enough speed for normal home/office networking.
Cat6	Zomato POS Terminal	It provides faster and more reliable wired communication for billing and order processing.
Fiber Optic	IPL Streaming Server	It supports very high-speed data over long distances with low signal loss, which is useful for live video streaming.

3. **Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.**

❖ Step-by-Step Process to Crimp an RJ-45 Connector

Aim: To attach an RJ-45 connector to a network cable and make an Ethernet cable.

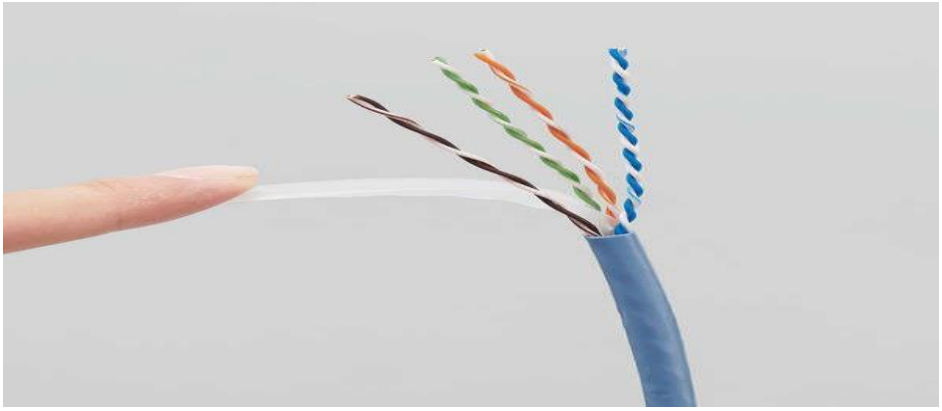
Tools required:

- Cat5e or Cat6 network cable
- RJ-45 connector
- Crimping tool
- Wire stripper or cable cutter
- Cable tester (optional)

➢ Step-by-step procedure

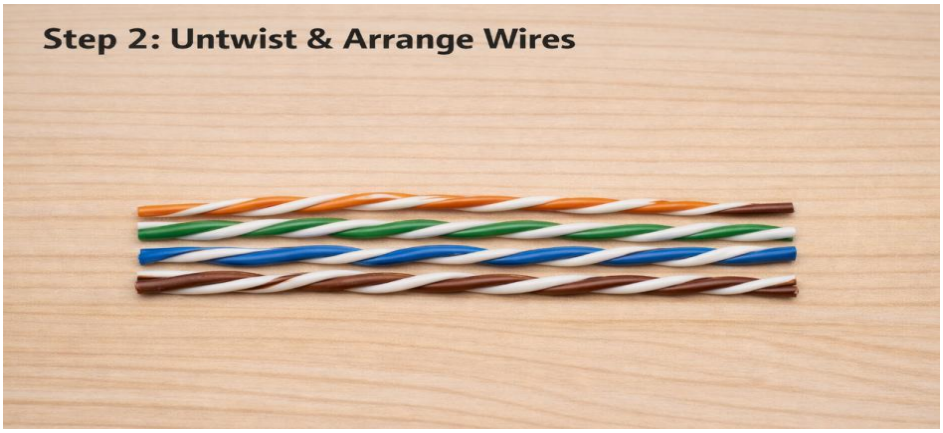
Reference video: How to Crimp RJ45 Connector – YouTube. The steps below are written in simple words based on the standard T568B method.

Step 1: Cut and strip the cable



Cut the network cable to the required length. Remove about 2–3 cm of the outer cover carefully without damaging the inner wires.

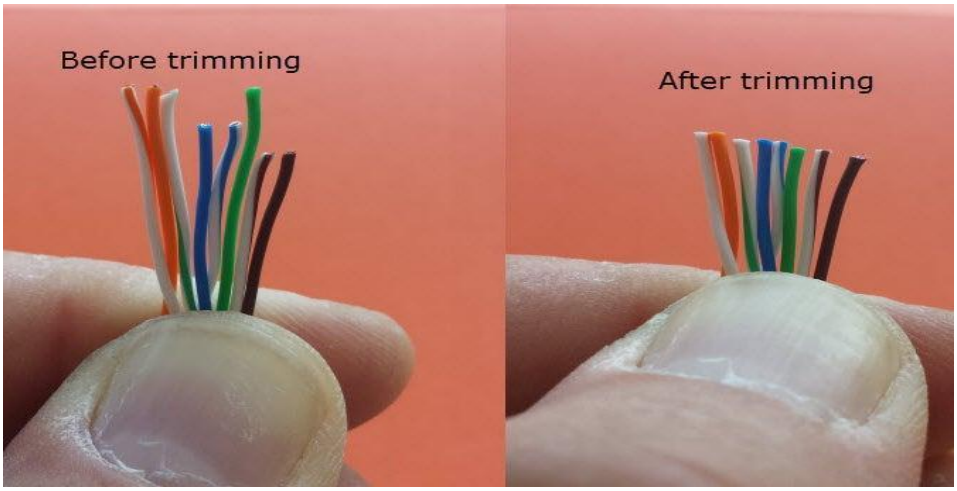
Step 2: Arrange the eight wires



Separate and straighten the eight wires. Arrange them in the T568B color order:

White-Orange, Orange, White-Green, Blue, White-Blue, Green, White-Brown, Brown.

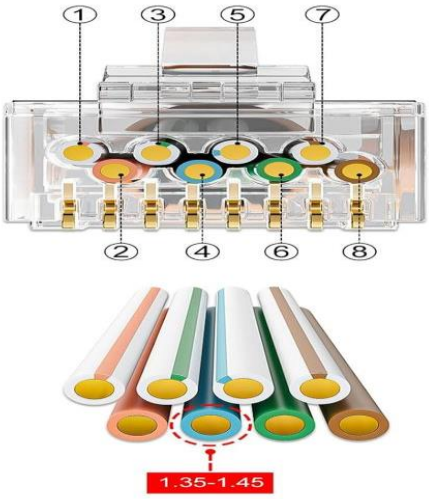
Step 3: Trim the wires



Hold all eight wires in the correct order and cut their ends evenly so that they fit properly inside the connector.

Step 4: Insert the RJ-45 connector





Push all eight wires into the connector. Check that every wire reaches the front and the outer cable jacket sits inside the connector.

Step 5: Crimp the connector



Place the connector in the crimping tool and squeeze the handles fully. This secures the wires and metal contacts in place.

Step 6: Test the cable



Repeat the process at the other end using the same T568B standard. Connect both ends to a cable tester and check that all eight wires are connected correctly.

➢ Two common mistakes to avoid

1. Wrong wire color order: If the wires are arranged incorrectly, the cable may not work properly. Always check the T568B color sequence before crimping.

2. Wires not inserted fully: If one or more wires do not reach the front of the connector, the connection may fail. Check all eight wires before crimping.

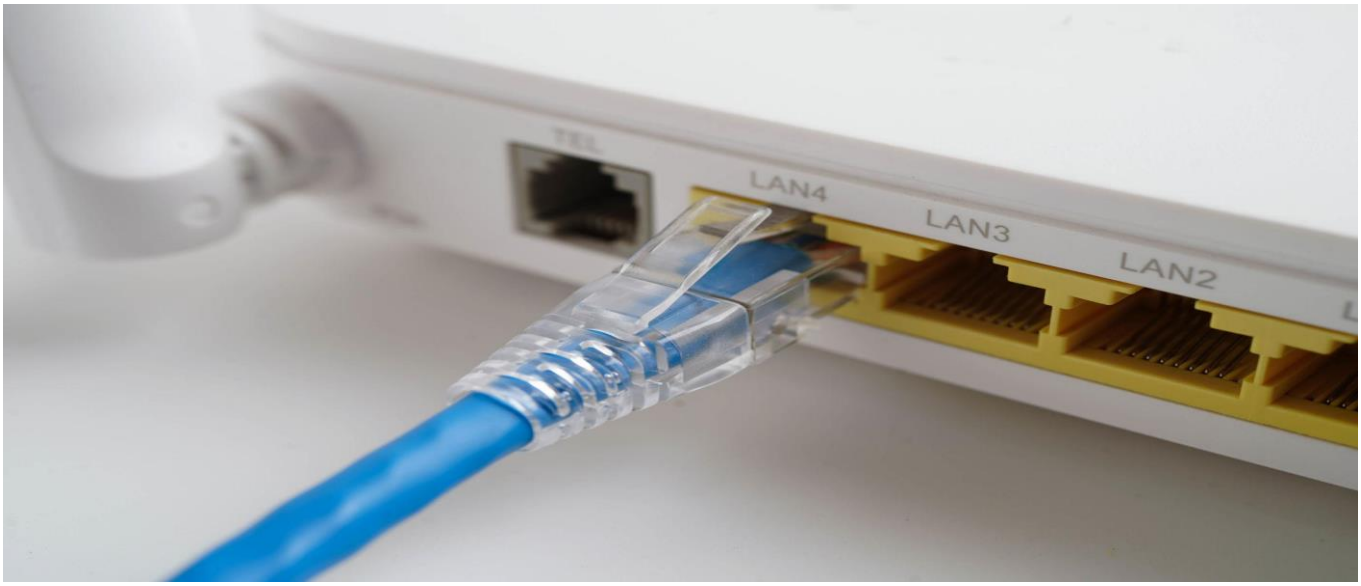
Conclusion: Crimping an RJ-45 connector correctly helps create a reliable Ethernet cable for connecting computers, routers, and switches

4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Cable Type	Connector Type	Usage	Reason
HDMI 2.1	HDMI connector	Connect gaming consoles to TVs or monitors	Supports high-quality video and audio with high refresh rates.
Cat6 Ethernet	RJ45 connector	Connect consoles, PCs, and network switches to the router	Provides fast, stable internet with low latency.
Fiber Optic	LC or SC connector	Connect network switches over long distances	Supports high-speed data transmission over long distances.
Power Cable	AC power plug	Supply electricity to TVs, consoles, and network devices	Provides power to all equipment.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.Hint: Look for blue or yellow cables with clear plastic ends (RJ-45) or thin round cables for fiber optics.

Photo: Ethernet Cable Connected to a Wi-Fi Router



Description :

- Cable type: Ethernet cable (LAN cable).
- Connector: RJ45.
- Devices connected: Wi-Fi router and computer or network switch.
- Use: It provides a fast and stable network connection for internet access.